

Overview

1 Random numbers

- Transformation method
- Rejection method

2 Monte Carlo integration

- Monte Carlo vs ordinary methods

3 Summary

Transformation method

Suppose X has probability distribution $p_X(x)$, take $Y = f(X)$.

The probability of finding X in $(x, x + dx)$ must be the same as the probability of finding Y in $(y, y + dy)$,

$$|p_Y(y)dy| = |p_X(x)dx|$$

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Since $p_X(x), p_Y(y)$ are both positive, this gives us

$$\begin{aligned} p_Y(y)|f'(x)dx| &= p_X(x)|dx| \\ \implies p_Y(y) &= \frac{p_X(x)}{|f'(x)|} \end{aligned}$$

Obtaining a specific distribution

We want a certain $p(y)$. What is $y = f(x)$ if x is uniform?

$$\begin{aligned}\frac{1}{f'(x)} = \frac{dx}{dy} = p(y) &\implies dx = p(y)dy \\ \implies x = \int_{-\infty}^y p(z)dz &\equiv \mathcal{C}(y)\end{aligned}$$

Inverting this gives us: $y(x) = \mathcal{C}^{-1}(x)$

x is uniformly distributed.

What transformation $y = f(x)$ will provide variable y with distribution $p(y)$?

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We can find the transformation function if

- 1 we can integrate our distribution $\rightarrow$ cumulative distribution $\mathcal{P}(y)$
- 2 we can invert the cumulative distribution function
analytically

Very useful for scaling and shifting distributions

Shifting and scaling

The transformation method is useful for shifting and scaling distributions:

$$Y = X/a + b \implies p_Y(y) = ap_X(x) = ap_X(a(y - b))$$

Example

X is gaussian with average 0 and variance 1.

We want Y to be gaussian with average μ and variance σ^2 ,

$$p_X(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(y-\mu)^2/2\sigma^2}$$

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We can understand this intuitively:

- Multiplying by σ stretches or squeezes the distribution
- Adding μ shifts everything to the left or right

Gaussian random numbers

We want a gaussian distribution

$$P(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

But

- we have no closed expression for $\int P(y) dy$
- we certainly have no closed expression for the inverse!

[scipy (in scipy.special) has functions `erf` and `erfinv` — numerical approximations]

A way around

We can integrate **two** gaussian distributions,

$$P(y_1, y_2) = \frac{1}{2\pi} e^{-(y_1^2 + y_2^2)/2} = P(y_1)P(y_2)$$

Gaussian random numbers

Transformation of two or more probability distributions:

$$P_Y(y_1, y_2, \dots) = P_X(x_1, x_2, \dots) \left| \frac{\partial(x_1, x_2, \dots)}{\partial(y_1, y_2, \dots)} \right|$$

If we take X_1, X_2 uniform on $\langle 0, 1 \rangle$ and

$$Y_1 = \sqrt{-2 \ln X_1} \cos(2\pi X_2) \quad Y_2 = \sqrt{-2 \ln X_1} \sin(2\pi X_2)$$

then Y_1, Y_2 are gaussian

Note

- Numpy has inbuilt gaussian prng — `np.random.randn()`
- Use the method above if you program in C/C++/Fortran/Java
- Avoid calls to `ln, sin, cos, ...` as much as possible: **they are slow!**

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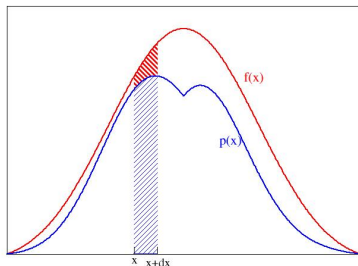
Rescale $f_0(x)$: $f(x) = Af_0(x)$, so that $f(x) > p(x)$ everywhere.

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Idea: Select points under $f(x)$ curve,
reject those in red shaded area

The ratio of areas is $p(x)/f(x)$

Rejection method

Implementation

- 1 Pick number X according to distribution $\frac{1}{A}f(x)$, where $A = \int_{-\infty}^{\infty} f(x)dx$
- 2 **Accept** X as your random number with probability $p(X)/f(X)$.
i.e., **reject** X with probability $p(X)/f(X)$.

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 - ▶ Use auxiliary random variable ξ : pick random uniform $\xi \in [0, 1]$
 - ▶ If $\xi < p(X)/f(X)$ then **accept** X as your random number
else **reject** X

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else **reject** X
- Store each accepted value in a list/array.
Issue: size of this array not known in advance?
- If $p(x)$ is normalised, $f(x)$ is *not* normalised; $\frac{1}{A}f(x) = f_0(x)$ is.
- Efficiency depends on A — choose A as small as possible while still satisfying $f(x) > p(x)$ everywhere.

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In physics we often have smallish perturbations on gaussian or exponential

Example

Generate a pseudo-random number with the distribution

$$p(x) \propto \frac{e^{-x}}{1+x^2}, \quad x > 0,$$

assuming we already have a generator for the exponential distribution.

Algorithm:

- 1 Generate an exponentially distributed number z .
- 2 Generate a standard uniform deviate u .
- 3 If $u < 1/(1+z^2)$, set $x = z$, otherwise go back to 1.
- 4 Repeat this to generate as many numbers x as you require.

Monte Carlo methods

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famous for its casinos

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Three main types

- Direct Monte Carlo
- Monte Carlo integration
- Markov chain Monte Carlo

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- Markov chain Monte Carlo
 - ▶ generate statistical distributions using 'random walks'
 - ▶ a stalwart of many-particle physics

Monte Carlo integration

We want to integrate a function f on interval $[a, b]$:

$$I = \int_a^b f(x) dx$$

Standard numerical integration (previous semester):

- divide $[a, b]$ into N evenly spaced points x_i
- evaluate $f(x)$ of those points

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Pick the points x_i randomly!

Monte Carlo integration

Sample randomly

$$I = (b - a) \times \langle f \rangle \approx \frac{b - a}{N} \sum_{i=0}^N f(\xi_i)$$

Where the ξ_i are uniformly sampled from $[a, b]$; i.e.,
are random numbers from distribution

$$p_{\xi}(x) = \begin{cases} 1/(b - a) & \text{if } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

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Advantages

- It works in arbitrary numbers of dimensions: $\int f dV = V \langle f \rangle$
- It works for complicated boundaries

Computing π

π is the area of the unit circle $x^2 + y^2 < 1$

$\frac{\pi}{4}$ is the area of the quadrant $x^2 + y^2 < 1; x, y \in \langle 0, 1 \rangle$

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This can be written as a 2-dimensional integral:

$$\begin{aligned}\frac{\pi}{4} &= \int_0^1 \int_0^1 \Theta(1 - (x^2 + y^2)) dx dy \\ &= \int_0^1 \int_0^1 \Theta(1 - (x^2 + y^2)) dx dy / \int_0^1 \int_0^1 dx dy\end{aligned}$$

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Procedure

- 1 Generate N pairs of random numbers (x, y)
- 2 Add 1 each time $x^2 + y^2 < 1$
- 3 Divide by N to get the average
- 4 Multiply by 4, and you have π !

Error estimates

The variance of a stochastic variable X is

$$\sigma_X^2 \equiv \text{var } X \equiv \langle (X - \bar{X})^2 \rangle = \langle X^2 \rangle - \langle X \rangle^2$$

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For our MC integral we get

$$\begin{aligned}\sigma^2 &= \left\langle \left(\frac{b-a}{N} \sum_i f_i \right)^2 \right\rangle - \left\langle \frac{b-a}{N} \sum_i f_i \right\rangle^2 \\&= \frac{(b-a)^2}{N} \left[\frac{1}{N} \left\langle \sum_i f_i^2 \right\rangle + \frac{1}{N} \left\langle \sum_{i \neq j} f_i f_j \right\rangle - N \langle f \rangle^2 \right] \\&= \frac{(b-a)^2}{N} \left(\langle f^2 \rangle + \frac{1}{N} \sum_{i \neq j} \langle f_i \rangle \langle f_j \rangle - N \bar{f}^2 \right) \\&= \frac{(b-a)^2}{N} \left(\langle f^2 \rangle + \frac{N(N-1)}{N} \langle f \rangle^2 - N \langle f \rangle^2 \right) = \frac{(b-a)^2}{N} \left(\langle f^2 \rangle - \langle f \rangle^2 \right)\end{aligned}$$

In going from the second to the third line we have used that f_i and f_j are independent and uncorrelated.

Advantages in many dimensions

In general:

$$\sigma^2 = \frac{V^2}{N} \left(\langle f^2 \rangle - \langle f \rangle^2 \right) \quad \text{— the error decreases as } \frac{1}{\sqrt{N}}$$

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Standard numerical integration gives errors $\propto$ powers of the grid spacing,

$$\text{err} \sim \delta^2 \sim \frac{1}{N}, \quad \delta^3 \sim \frac{1}{N^3}, \dots$$

Adding more points gives much faster improvement in accuracy than MC

Not good for Monte Carlo in one dimension, **but...**

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In d dimensions $N \sim (\frac{L}{\delta})^d$ so $\text{err} \sim N^{-k/d}$, not so fast for large d

The Monte Carlo error is **still** $\text{err} \sim N^{-1/2}$

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Monte Carlo integration

- Integrate functions by randomly sampling points
- Errors decrease as $1/\sqrt{N}$
- Superior for high-dimensional integrals